Example 7.4

A person deposits ₹4,000 in the beginning of every year. If the rate of compound interest is 14% then ,find the amount after 10 years . $[(1.14)^{10}=3.707]$.

Solution :

Here $a = 4{,}000$; $i = 0.14$ and $n = 10$.

$A = (1+i)\ \frac{a}{i}[(1+i)^n-1]$

$= (1+0.14)\ \frac{4000}{0.14}\ [(1+0.14)^{10}-1]$

$= (1.14)\frac{4000}{0.14}\ [(1.14)^{10}-1]$

$= 1.14\times\frac{4000}{0.14}(3.707-1)$

$= 1.14\times\frac{4000}{0.14}(2.707) = 88{,}170.86$

$\therefore\ A \approx$ ₹88,171

Example 7.5

A person purchases a machine on 1st January 2009 and agrees to pay 10 installments each of ₹12,000 at the end of every year inclusive of compound rate of 15%. Find the present value of the machine. $[(1.15)^{10}=4.016]$.

Solution:

Here $n=10$, $a = 12{,}000$ and $i = 0.15$

Now $P = \frac{a}{i}\left[1-\frac{1}{(1+i)^n}\right]$

$= \frac{12{,}000}{0.15}\left[1-\frac{1}{(1+0.15)^{10}}\right]$

$= \frac{12{,}000}{0.15}\left[1-\frac{1}{(1.15)^{10}}\right]$

$= \frac{12{,}00{,}000}{15}\left[1-\frac{1}{4.016}\right]$

$= 80{,}000\left[\frac{4.016-1}{4.016}\right]$

$= 80{,}000\left[\frac{3.016}{4.016}\right] \approx 60{,}080$

$\therefore\ P =$ ₹ 60,080

Example 7.6

A photographer purchases a camera on installments. He has to pay 7 annual installments each of ₹36,000 right from the date of purchase. If the rate of compound interest is 16% then find the cost price (present value) of the camera. $[(1.16)^7=2.828]$

Solution:

Here $a = 36{,}000$; $n = 7$ and $i = 0.16$

Now $P = (1+i)\ \frac{a}{i}\left[1-\frac{1}{(1+i)^n}\right]$

$= (1+0.16)\ \frac{36{,}000}{0.16}\ \left[1-\frac{1}{(1+0.16)^7}\right]$

$= \frac{1.160}{0.16}(36{,}000)\left[1-\frac{1}{(1+0.16)^7}\right]$

$= \frac{116\times 36000}{16}\ \left[1-\frac{1}{2.828}\right]$

$= \frac{116\times 36000}{16}\ \left[\frac{2.828-1}{2.828}\right]$

$= \frac{116\times 36{,}000\times 1.828}{16\times 2.828}$

$= \frac{116\times 36{,}000\times 1828}{16\times 2828} \approx 1{,}68{,}709.$

$\therefore$ P = ₹ 1,68,709

Example 7.7

A person has taken a loan of ₹ 7,00,000 at 16% rate of interest from a finance company. If the repayment period is of 15 years then find the installment he has to pay at the beginning of each month. $[(1.0133)^{180}=9.772]$.

Solution:

Here $P = 7{,}00{,}000$; $n = 15$; $i = 0.16$ and $k = 12$.

$P = \left(1+\frac{i}{k}\right)\left(\frac{a}{\frac{i}{k}}\right)\ \left[1-\frac{1}{\left(1+\frac{i}{k}\right)^{nk}}\right]$

$700000=\left(1+\frac{0.16}{12}\right)\left(\frac{a}{\frac{0.16}{12}}\right)\left[1-\frac{1}{\left(1+\frac{0.16}{12}\right)^{15\times 12}}\right]$

$= \left(\frac{12.16}{12}\right)\left(\frac{12a}{0.16}\right)\left[1-\frac{1}{\left(\frac{12.16}{12}\right)^{180}}\right]$

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